Imperial College
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# Lecture 5

# Applications of Operational Amplifiers

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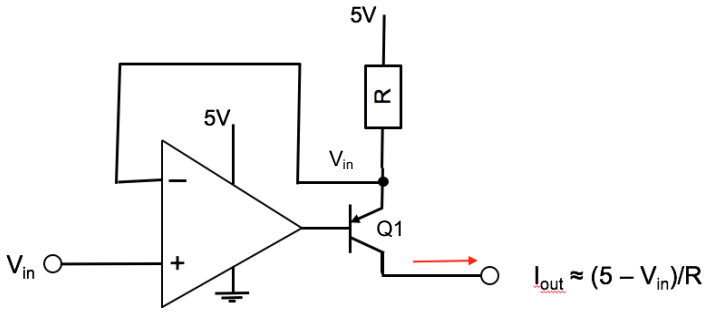
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In lecture, I will be exploring a number of applications using op-amp as the building block. These are commonly used circuits, some of which you have already come across in Year 1, either in the Analysis and Design of Circuits module or in the Summer term group project.

To support this lecture, you will also be building and testing most of the circuits from this lecture in Laboratory Experiment 2 this week.

# Voltage to current converter




- PNP transistor Q1 must be in linear region
- Op-amp forces V- = Vin
- R (with 5V) determines the current as $I_R = (5 - V_{in})/R$
- Assume no current flows into input of op-amp
- $I_C = I_E - I_B$, assume current gain $\beta >> 1$, $I_C \approx I_E \approx I_R$
- Can use FET or MOSFET in place of BJT



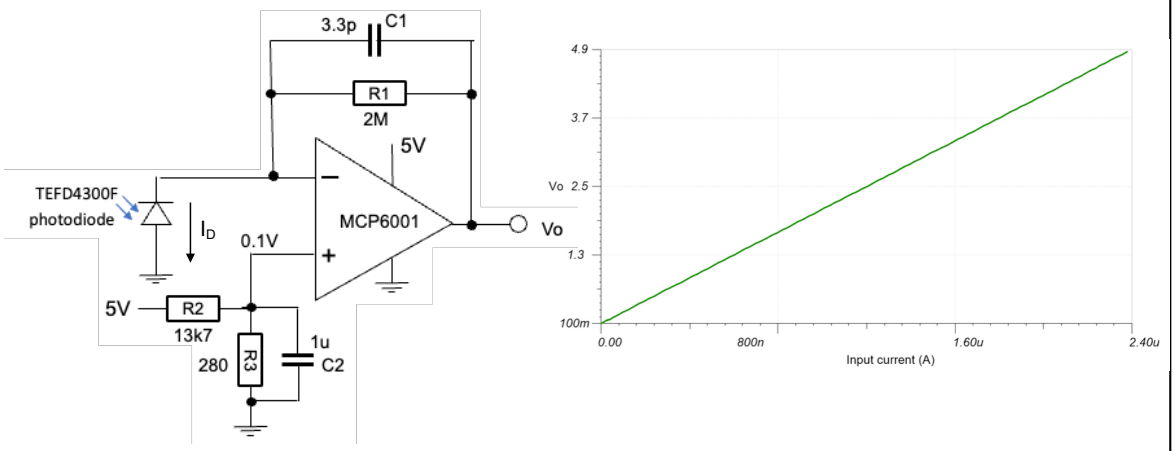
We can convert a voltage $V_{in}$ to a current $I_{out}$ that is proportional to the voltage with high accuracy using this circuit. The PNP transistor Q1 MUST BE working in the linear region at all times. The op-amp's negative feedback forces the voltage at V- input pin to $V_{in}$. The resistor R determines the current flowing into the emitter of Q1.

Assuming that the current gain of Q1 is relatively large (say at least 100), then $I_{out} = I_C \approx I_R$. The output current is independent of the collector voltage of Q1. Therefore, this is a good current source.

Instead of using a BJT, one could use either a p-channel FET or MOSFET.

There is however one important condition: since the current must flow OUT of this current source, the load must be connected to ground or a negative supply.

## Current amplifier (photo detector)




- Photodiode generates reverse current proportional to light intensity (visible or IR) received.
- Inverting op-amp used to convert diode current into a voltage.
- R2 and R3 provides an offset at the output when photodiode current is 0.



So far, we have been using op-amps to amplify voltages. In the case of photo sensitive diodes, the input signal is a **current** that is dependent on the light fallen onto the device.

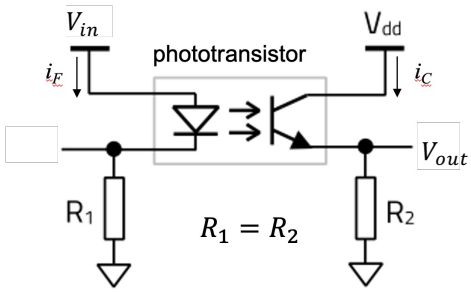
Amplifying such a current source is simple – we use an inverting amplifier configuration where the V- input is a virtue 'earth' at a nominal ground potential. The negative feedback through R1 and C1 ensures that the diode current generated by the photons produces a voltage at the output:

$$V_0 = I_D R_1.$$

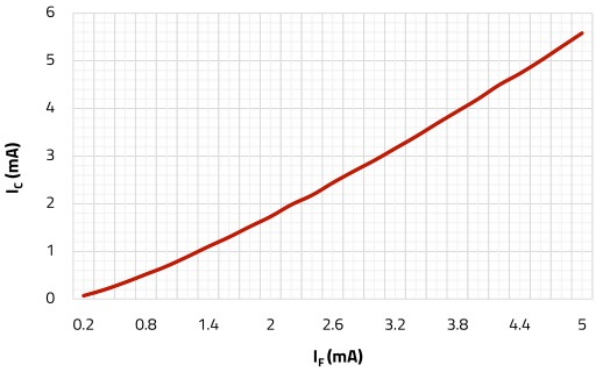
Note that the diode current is a reverse leakage current through the diode, and therefore it is flowing in the direction shown.

The V+ input of the op-amp is biased to be at around 100mV so that $V_O$ is 100mV even when the photodiode produces NO current. This is to avoid hitting the non-operational region of the output since minimum output voltage is, say, 25mV above Vss.

# Optical isolating amplifier




Current Transfer Ratio (CTR) = $\frac{i_c}{i_F}$




- Phototransistor provides electrical isolation (e.g. 5kV) between two parts of circuit.
- Consist of a light emitting diode and a photosensitive transistor.
- Diode current, through light, is transferred to transistor output current (current transfer ratio, CTR).
- The relationship between the diode current $i_F$ and transistor current $i_C$ is **not** linear.



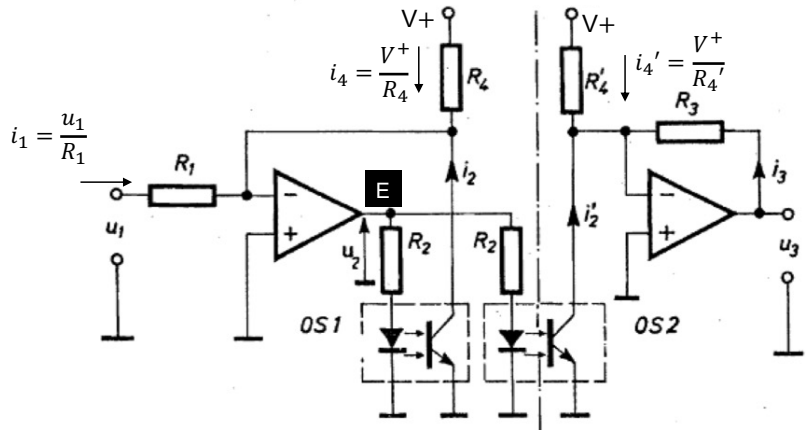
A phototransistor consists of a light emitting diode and a photo sensitive BJT whose collector current is approximately (but not exactly) proportional to the light received. The two devices are housed in the same package.

When a current $I_F$ flows through the diode, a corresponding current $I_c$ flows through into the collector of the transistor as shown above. The ratio of $\frac{I_C}{I_F}$ is known as the **current transfer ratio (CTR)**, and it is roughly 1 (but not exactly). This is shown in the graph above.

Since there is NO physical connection between the left and right side of the circuit, this arrangement serves as an isolation stage. For example, in biomedical devices, it is absolutely necessary that the circuits that is connected to a human body is isolated from the main power supply to avoid harming the person.

Such a device typically provides a few to a few tens of kV isolation between the two sides of the circuit.

## Optical isolating amplifier




- To mitigate the non-linear characteristics of the phototransistor, use two "matching" phototransistors as shown above.
- Output of 1st op-amp $u_2$ settles to a value such that KCL holds true at the virtual earth node E. That is $(i_1 + i_2 + i_4) = 0$.
- Assuming $R_4 = R_4'$, then $i_2' = i_2$. The symmetry makes $i_3 = i_1$.
- Hence, $u_3 = i_3' \times R_3 = \frac{R_3}{R_1} \times u_1$.



Since the current transfer ratio (CTR) is not exactly 1, a circuit is needed to ensure the two sides of the circuit are linear and proportional. This can be achieved with two op-amps designed as current amplifiers as shown above.

The input current $i_1 = \frac{u_1}{R_1}$ is forced by the left op-amp into the left phototransistor OS1. Since V- input is at virtue earth, the current through $R_4$ is constant ($i_4 = V^+/R_4$). This ensures that the output voltage of left op-amp is at a voltage that makes $i_1 + i_4 = -i_2$.

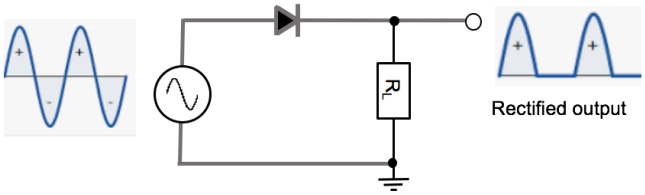
Since the right phototransistor OS2 is also driven by the same voltage from the op-amp, the output current of OS2 $i_2' = i_2$. We assume here that the two phototransistors are matching.

Due to the high input impedance of the op-amp and constant current through $R_4'$, the collector current $i_2'$ has to flow through the feedback resistor $R_3$. Therefore, the output voltage:

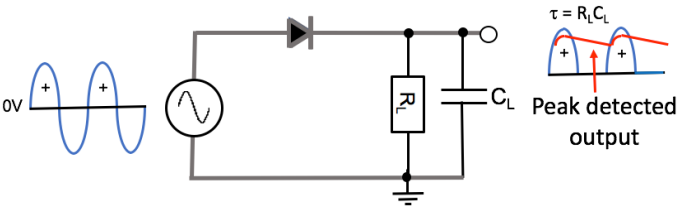
$$u_3 = \frac{R_3}{R_1} u_i.$$

This is therefore a non-inverting amplifier as the case of other op-amps, but the input and output sides of the circuit are completely isolated up to a few kilo volts.

## Half-wave rectifier & Peak detector




- Diode and resistor – simple half-wave rectifier
- Commonly used in power electronics or and multimeters




- $C_L$ charges to $V_{in}$ peak – $V_D$
- Diode prevents $C_L$ discharging when $V_{in}$ drops
- $R_L$ discharges capacitor with time constant $R_LC_L$
- CL charges again on the positive cycle

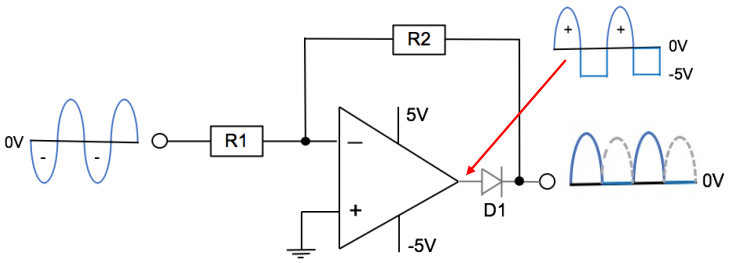
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You are familiar with how diodes work. Here is a simple rectifier circuit using a single diode. Only the positive going half cycle of the input sine wave will forward bias the diode for current to flow from the source to $R_L$. On the negative half cycle, the diode is reverse biased and no current (except leakage current) flows to the load.

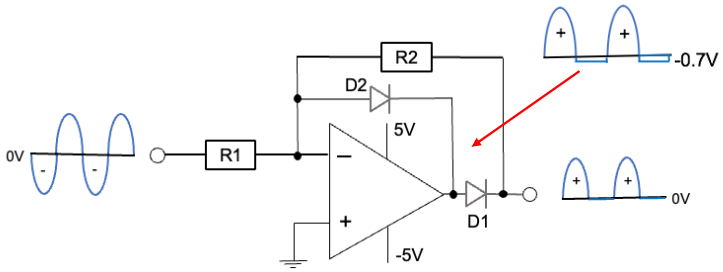
Such rectifier circuit is commonly used in power supply circuits to convert 50Hz (60Hz in North America) mains voltage to DC. You will learn more about such power conversion circuits next term on the Power Electronics module. You will also find similar circuit in a multimeter, which converts the measured ac signal to dc either through averaging (lowpass filtering) or through peak detection.

Adding a capacitor in parallel to $R_L$ results in a peak detector circuit. The capacitor charges to when $V_{in} \geq V_C + V_D$ through the diode which is forward biased. When $V_{in} < V_C + V_D$ , the diode is no longer forward biased. The capacitor discharges through the load resistors $R_L$. This peak detector circuit produces an output that is roughly a DC voltage but it contains ripple. The size of theripple is dependent on the time constant $R_LC_L$ and the frequency of the input signal. For the peak detector to work effective $R_LC_L$ >> period of the signal (e.g. $R_LC_L$ = 10 / Fs, where Fs is the signal frequency).

## Rectifier with op-amp buffering




- Assume R1 = R2
- Negative cycles result in an inverting amplifier with gain = -1
- Op-amp drives output with low impedance
- Positive cycles, op-amp isolated from output
- Poor full-wave rectification




- D1 provides feedback path for negative input cycles
- D2 provides feedback path for positive input cycles
- Op-amp operating throughout entire cycle

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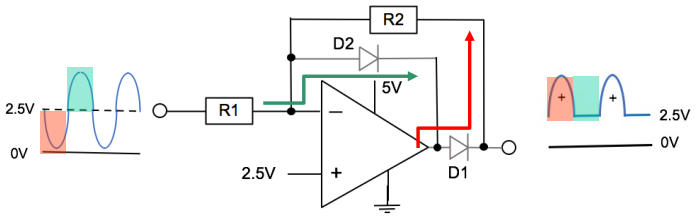
Op-amp can be used to implement a rectifier circuit. Let us assume that we are using a dual ±5V supply op-amp, and the reference voltage is GND.

Top circuit uses only one diode. The negative half cycle of the input forces the op-amp output to go positive, forward biasing D1 to complete the feedback loop. This results in a low impedance drive to the output with a positive voltage as shown.

On the positive half cycle of the input, the op-amp output goes negative reverse biasing D1. Now the feedback loop is broken by the diode. The op-amp output goes to -5V. The V- input is no longer virtual earth at GND potential. The output is now driven through R1 and R2. The voltage at the output will be the same as the positive input signal, but subject to the loading effect.

A somewhat better circuit is shown below. Here we add another diode D2. This provides feedback path for the positive half of the input signal. The result is a half-wave rectifier with the output always driven by the op-amp whose V- input is forced to be at virtual earth GND voltage.

## Single power supply “rectifier”

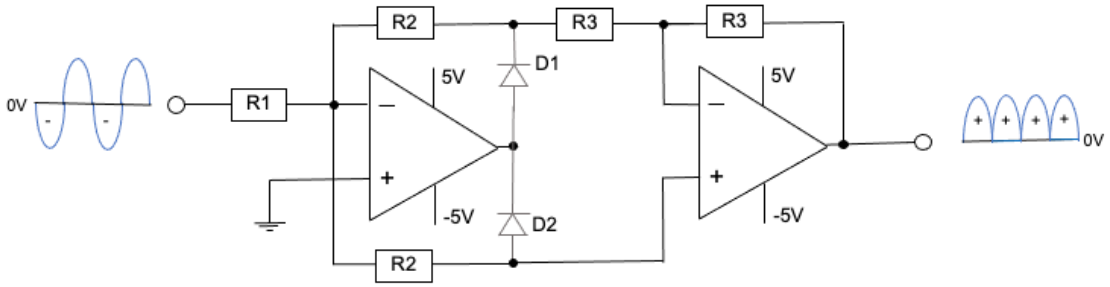



- Single power supply rectifier is implemented by shifting the reference voltage to ½ $V_{DD}$



Single supply op-amp can also be used as rectifier. Of course “rectification” with only one supply is a bit unusual. Here we lift the reference voltage from 0V to 2.5V (for VDD = 5V), and everything will work as before. However, both input and output are now relative to the 2.5V offset.

# Full-wave rectifier




- Precision full-wave rectifier with two op-amps
- Op-amp 1 provides two separate half of the rectified signals
- Op-amp 2 sums two half cycles together



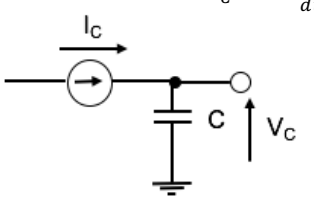
A full-wave rectifier that can drive an output load effectively can be implemented with two op-amps.

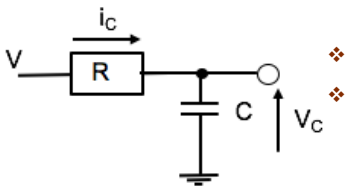
The two paths through D1 and D2 provide negative feedback for negative and positive half cycle of the input respectively.

The second op-amp acts as a summing circuit that adds the two half together to provide a full-wave rectified output.

# Integrator

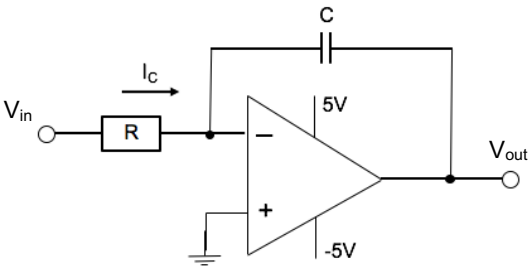
- $i_C = C\frac{dV_C}{dt} \Rightarrow V_C = \frac{1}{C}\int i_C dt + V_C(0)$

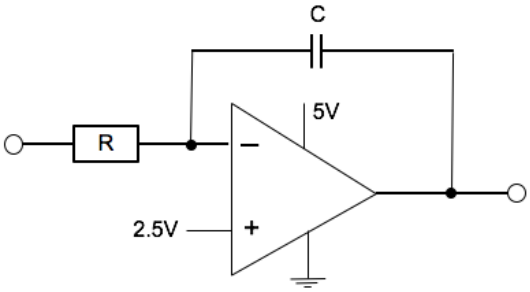



- $I_C$ changes with $V_C$
- $V_C$ is an exponential rise function (not perfect integral)

- Constant $I_C$, $V_C = \frac{i_C}{C}t + V_C(0)$




- $v_{out} = -\frac{V_{in}}{RC}t + V_C(0)$
- Single supply operation



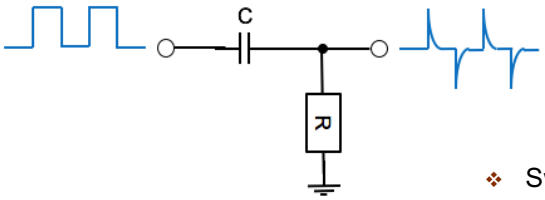
If we have an ideal constant current source, a capacitor will integrate the current perfectly to produce Vc(t). Vc(0) is the initial capacitor voltage.

One could use a resistor to convert voltage V to a current to charge the capacitor. This of course will not produce a very good integrator because the current charging the capacitor is no longer constant. As the capacitor charges up, Vc increases and Ic decreases. Vc(t) follows the exponential rise function.

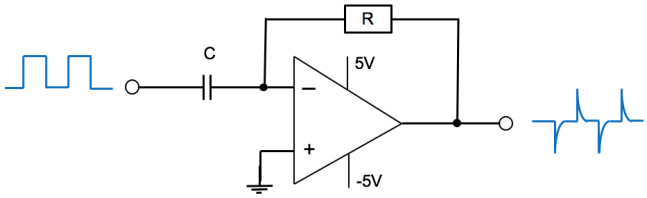
We can implement a near perfect integrator using an op-amp as shown on the slide. The inverting op-amp has V- node fixed as virtual earth, and R is now used to converter $V_{in}$ to a current $I_C = V_{in}/R$. The current has no where to go except to charge capacitor C.

Implementing an op-amp integrator using single power supply is again straight forward. The virtual earth node is now a virtual 2.5V, which is the reference voltage for the circuit.

# Differentiator




- Swap R and C
- Implement a differentiator
- Not used often because circuit tends to produce noisy output



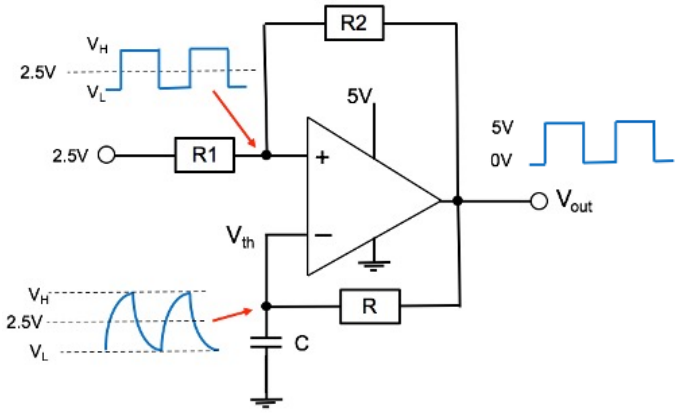



Swapping R and C in both passive and op-amp circuit for the two circuits as integrators yields two differentiator circuits. Both provide an approximation to the differentiation function as shown.

The way it works is that the voltage across a capacitor cannot change instantaneously. Therefore, the output follows the change in input before the capacitor charges (or discharges).

Differentiator circuits are not popular. It tends to amplify high frequency signals and produces a very noisy output which makes them not useful in practical applications.

# Simple Oscillator




- Combine comparator with hysteresis and RC network = oscillator
- Voltage at V+ change instantly with $V_{out}$:

$$V_H = 2.5\left(1 + \frac{R1}{R1+R2}\right)$$

$$V_L = 2.5\left(\frac{R2}{R1+R2}\right)$$

- V- = $V_{th}$ rises and falls exponentially with a time constant of RC between $V_H$ and $V_L$
- This is determined by the equation:

$$V_{th} = V_f + (V_i - V_f)e^{-\frac{t}{\tau}}$$

$V_i$ = initial value, $V_f$ = final value, $\tau$ = time constant RC



This shows a simple square wave oscillator circuit. Assume $V_{out}$ is oscillating between $V_{DD}$ = 5V and $V_{SS}$ = 0V. For the given circuit, V+ node voltage changes instantly with Vout between $V_H$ and $V_L$ as shown (simple voltage divider).

However, V- node cannot change instantaneously because of C. Instead, the voltage $V_{th}$ follows an exponential rise and fall equation.

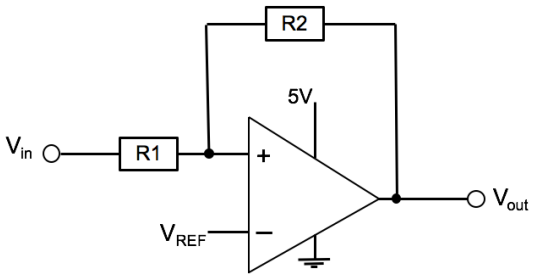
In general, for a C that is charged or discharged from a voltage source through a resistor R, the capacitor voltage is given by the equation:

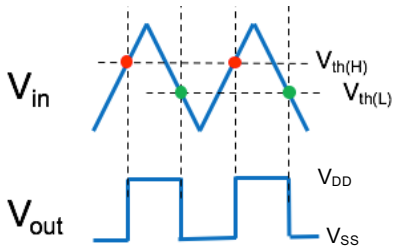
$$V_C = V_f + (V_i - V_f)e^{-\frac{t}{\tau}}$$

where $V_i$ and $V_f$ are the initial and final values of the voltage $V_C$ respectively and $\tau$ is the time constant RC.

For the rising portion of $V_{th}$=$V_C$, $V_i$ = $V_L$ and $V_f$ = 5. For the fall portion of $V_{th}$ , $V_i$ = $V_H$ and $V_f$ = 0.

## Comparator with hysteresis




- $V_{out}$ swings between $V_{DD}$ and $V_{SS}$
- $V_{out}$ changes state when V+ reaches $V_{REF}$
- Apply KCL at V+:

$$\frac{V_{REF}-V_{in}}{R1}=\frac{V_{out}-V_{REF}}{R2}$$

$$\Rightarrow V_{in}=V_{REF}\left(1+\frac{R1}{R2}\right)-V_{out}\left(\frac{R1}{R2}\right)$$

- If R1 = 0 or R2 = ∞, $V_{th} = V_{REF}$
- R1 > 0, R2 ≠ ∞

$$V_{th(H)}=V_{REF}\left(1+\frac{R1}{R2}\right)-V_{SS}\left(\frac{R1}{R2}\right)$$

$$V_{th(L)}=V_{REF}\left(1+\frac{R1}{R2}\right)-V_{DD}\left(\frac{R1}{R2}\right)$$

- Hysteresis = $V_{th(H)}-V_{th(L)}$



Op-amp can be used as an analogue comparator. In the circuit shown, $V_{REF}$ is constant voltage that defines the comparator threshold for switching output from high ($V_{DD}$) to low ($V_{SS}$) voltages. $V_{out}$ changes state when V+ reaches $V_{REF}$ from either direction (i.e. with $V_{in}$ rising or falling).

Applying KCL at V+ node yields the equation for $V_{in}$ when switching occurs:

$$V_{th}=V_{REF}\left(1+\frac{R1}{R2}\right)-V_{out}\left(\frac{R1}{R2}\right)$$

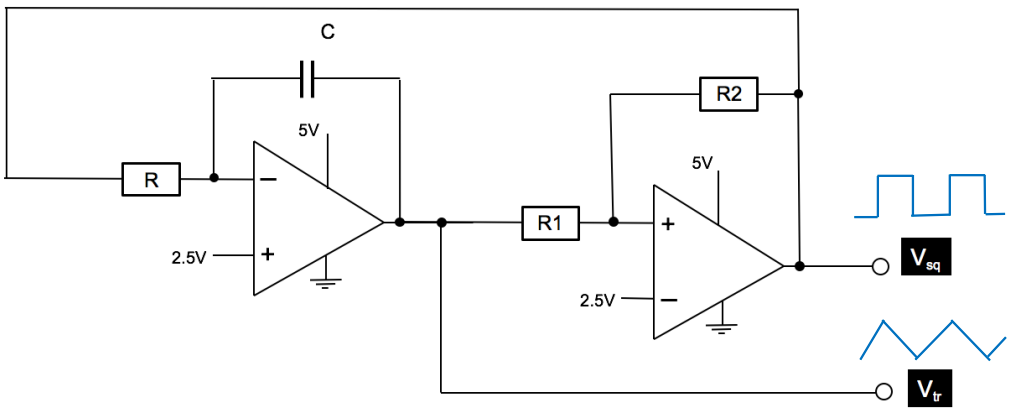
If R1 = 0 or R2 is open circuit, the $V_{th} = V_{REF}$.

It the op-amp has high gain-bandwidth product and high output slew rate (i.e. the maximum rate of change of the output voltage), the switching condition of $V_{th} = V_{REF}$ could results in output oscillation if $V_{in}$ is changing slowly.

With R1 > 0 and R2 < ∞ , the switching threshold is dependent on the state of $V_{out}$. Therefore the switching threshold when the signal is rising is different from that when the signal is falling. This create the hysteresis effect.

This circuit is also known as a **Schmitt trigger** circuit.

## Triangular and Square wave generator



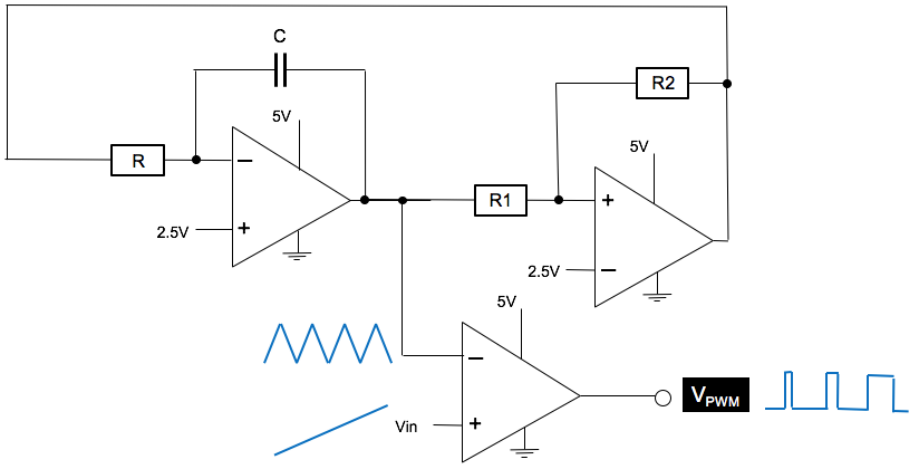

- Better oscillator circuit using integrator + comparator with hysteresis
- Integrator output produces a triangular signal
- Comparator (with hysteresis) produces a square signal
- Feedback circuit ensures oscillation is maintained



This oscillator generates both a triangular signal and a square signal at the same time. It uses an op-amp integrator to produce a negative going ramp when the integrator input is at $V_{DD}$. It produces a positive going ramp when the integrator input is at $V_{SS}$ or 0V.

Since the comparator is designed to have hysteresis, the triangular signal causes the comparator to switch state when $V_{th(H)}$ and $V_{th(L)}$ are reached. (See slide 13.)

## Pulse-width Modulator




- Comparing triangular signal with $V_{in}$ -> pulse-width modulated output
- Frequency of triangular signal >> $V_{in}$ frequency
- Output pulse width proportional to value of $V_{in}$
- Recover $V_{in}$ by lowpass filtering $V_{PWM}$

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We can implement a pulse-width modulator by simply comparing the input signal Vin to the periodic triangular signal. The output is a sequence of pulses whose widths are proportional to the value of Vin. For this to work, the frequency of the triangular signal must be much higher than that of the input signal Vin.

You will find PWM circuits in many devices, particularly in microcontrollers. We will also be examining how to produce PWM signals in digital circuits later in this module.

Pulse-width modulated signal is very useful in power electronics where we want to obtain an average voltage through switching transistors ON and OFF. PWM signal is also commonly used to control speed of motors. When a transistor (BJT, FET or MOSFET) is used as a switch (instead of a linear device), the the switch resistance is low when the current is high during the ON state, and the resistance is very high, but the current is low during the OFF state. Therefore, the power dissipated (or wasted) by a transistor controlled by a PWM signal is low. This leads to higher efficient in the system.